

Coherent Springer theory

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Lecture 1. G/F split, $T \subset B \subset G$

$\text{Rep}(G)$ derived co-cat of smooth reps of $G(F)/\overline{G}_e$

Par_G moduli stack of Langlands parameters $/\overline{G}_e$

$\{(p, N) : p : W_F \rightarrow \hat{G} \text{ open kernel, } N \in \text{Lie}(\hat{G}), \text{Ad}_p(w) \cdot N = (w|N)\} / \hat{G}$

(u, ψ) Whittaker datum, $\psi : U \rightarrow \overline{G}_e^{\times}$ generic

$$W_{u, \psi} = \text{Ind}_U^G \psi.$$

Conj: $\exists$ fully faithful functor $\text{CLLC}_G : \text{Rep}(G) \rightarrow \text{IndCoh}(\text{Par}_G)$ s.t.

1) $\forall P \subseteq MU$ std

$$\begin{array}{ccc} \text{Rep}(M) & \xrightarrow{\text{CLLC}_M} & \text{IndCoh}(\text{Par}_M) \\ \downarrow \text{I}_{P \circ P}^G & & \downarrow q^* \\ & & \text{IndCoh}(\text{Par}_P) \\ & & \downarrow p_* \\ \text{Rep}(G) & \xrightarrow{\text{CLLC}_G} & \text{IndCoh}(\text{Par}_G) \end{array} \quad \text{Par}_M \xleftarrow{q} \text{Par}_P \xrightarrow{p} \text{Par}_G$$

2) $\text{CLLC}_G(W_{u, \psi}) = \mathcal{O}_{\text{Par}_G}$

$$\text{Rep}(G)_{[T, 1]} \quad \text{Par}_G^1 = \{(F, N) : \text{Ad}_F \cdot N = qN\} / \hat{G}$$

$$\downarrow \\ \text{I}_{B \circ P}^G \chi^{\text{univ}}$$

$$\mathcal{S}_G^1 = p_* \mathcal{O}_{\text{Par}_B^2}$$

$$\text{Ends} = H_G^{\text{aff}}$$

$$\text{CLLC}_G(\text{I}_{B \circ P}^G \chi^{\text{univ}})$$

If $H_G^{\text{aff}} = \text{End}(S_G^1)$, then

$V \mapsto \text{Hom}(I_{B^{\text{op}}}^G \chi^{\text{univ}}, V) \otimes_{H_G^{\text{aff}}} S_G^1$ is fully faithful

$$\text{Rep}(G)_{[T, 1]} \longrightarrow \text{IndCoh}(\text{Par}_G^1)$$

Th (BECHN): $\forall$ split G , $\text{End}(S_G^1) \simeq H_G^{\text{aff}}$ compatibly with 1) and 2),

so conj. holds on $\text{Rep}(G)_{[T, 1]}$. Moreover if $G = GL_n$, conjecture holds on all of $\text{Rep}(G)$.

Step 0: Basic reductions

0a: suffices to prove for G adjoint.

0b: replace H_G^{aff} with $H_{G,t}^{\text{aff}}$:

H_G^{aff} has a presentation with generators T_S

relations $(T_S - q)(T_S + 1)$

braid relations

$H_{G,t}^{\text{aff}}$ is a $\overline{\text{UE}}[t]$ -alg
has generators T_S

relations $(T_S - t)(T_S + 1)$

braid relations

$$\text{Par}_G^1 \rightsquigarrow \text{Par}_{G,t}^1 = \{(F, N, t): \text{Ad}(F)N = tN\}$$

$$S_{G,t}^1 = p_* \mathcal{O}_{\text{Par}_{B,t}^1}$$

Suff. to show $\text{End}(S_{G,t}^1) \simeq H_{G,t}^{\text{aff}}$

$$t \rightsquigarrow q$$

Step 1: $\hat{B}' = \hat{B} \times G_m, \quad \hat{G}' = \hat{G} \times G_m$

Springer resolution $N_{\hat{B}/\hat{B}'} \longrightarrow N_{\hat{G}'/\hat{G}'}$

$$St_{\hat{G}} = N_{\hat{B}/\hat{B}'} \times_{N_{\hat{G}'/\hat{G}'}} N_{\hat{B}/\hat{B}'}$$

Th (Kazhdan - Lusztig) $H_{G,t}^{alt} \simeq K_0(St_{\hat{G}})$

Step 2: $K(St_{\hat{G}})$ connective K-theory spectrum of $St_{\hat{G}}$

ring spectrum in char 0 $\rightsquigarrow$ dg algebras.

$$K(St_{\hat{G}}) \longrightarrow HH_*(St_{\hat{G}})$$

$$\downarrow \quad \nearrow$$

$$K_0(St_{\hat{G}}) \quad \text{cho}$$

Th (BZCHN): $HH_*(St_{\hat{G}})$ is conc. in deg 0. In particular, cho exists, and is a quasi-isomorphism.

Pt. $HH_*(St_{\hat{G}}) = HH_*(IndCh(St_{\hat{G}}))$

$$\text{Set } \overline{St_{\hat{G}}} = N_{\hat{B}/\hat{B}} \times_{N_{\hat{G}'/\hat{G}'}} N_{\hat{B}/\hat{B}}$$

Th (Berglund-Mirkov) $Shv(L^+I \backslash LG/L^+I, \overline{Q}_e) \simeq IndCh(\overline{St_{\hat{G}}})$.

$$HH_*(\overline{St_{\hat{G}}}) \simeq HH_*(Shv(L^+I \backslash LG/L^+I, \overline{Q}_e))$$

$L^+I \backslash LG/L^+I$ has stratification. $Shv(L^+I \backslash LG/L^+I)$ has semi-orthogonal decomp where its summands are $\langle \Delta_w \rangle$.

$$HH_*(\overline{St}_{\hat{G}}) = \bigoplus_w HH_*(\langle \Delta_w \rangle) \quad \text{lift } \langle \Delta_w \rangle \text{ from } \text{Coh}(\overline{St}_{\hat{G}})$$

to $\text{Coh}(St_{\hat{G}'})$, $\tilde{\Delta}_w$

$$HH_*(St_{\hat{G}'}) = \bigoplus HH_*(\langle \tilde{\Delta}_w \rangle)^{G_m}$$

$$\begin{cases} 0 & \text{in deg} > 0 \\ \overline{G}_0[t] & \text{in deg } 0 \end{cases}$$

Upshot: $H_{G,t}^{\text{aff}} \simeq HH_*(St_{\hat{G}'})$

Step 3: Loop spaces

$$X \text{ nice (derived) stack / } \overline{G}_e, \quad LX = X \times_{X \times X} X$$

$$X = * / \hat{G}, \quad LX = \hat{G} / \hat{G}$$

$$X = N_{\hat{G}} / \hat{G}' = \hat{G} \times G_m$$

$$LX = \left\{ (N, F, t): \begin{matrix} \text{Ad}(F) \cdot N = tN \\ \text{Par}_{\hat{G}', t} \end{matrix} \right\} / \hat{G}'$$

$$X = St_{\hat{G}'} \quad , \quad LX = \text{Par}_{\hat{G}', t}^1 \times_{\text{Par}_{\hat{G}', t}^1} \text{Par}_{\hat{G}', t}^1$$

$$f: X \rightarrow Y \quad \text{End}(f^* \mathcal{O}_X) \quad X_Y^X \rightarrow X$$

$$\begin{matrix} \text{is} \\ \Gamma(X_Y^X, \omega_{X_Y^X}) \end{matrix} \quad \begin{matrix} \downarrow & \downarrow \\ X & \rightarrow Y \end{matrix}$$

Lecture 2 Goal: $\text{End}(S^1_A) \simeq H^{\text{aff}}_A$, suff. to show $\text{End}(S^1_{G,t}) \simeq H^{\text{aff}}_{G,t}$

$$S^1_{G,t} : P \times \mathcal{O} \text{Par}^1_{B,t} \quad , \quad \text{Par}^1_{B,t} \xrightarrow{P} \text{Par}^1_{G,t}$$

$$H^{\text{aff}}_{G,t} \simeq \text{HH}_*(\text{St}^1_{\hat{G}'}) \quad , \quad \text{St}^1_{\hat{G}'} = \left(N_{\hat{B}} / \hat{B}' \right) \times_{N_{\hat{G}} / \hat{G}'} \left(N_{\hat{B}} / \hat{B}' \right)$$

$$LX = X \times_{X \times X} X$$

$$\hat{G}' = \hat{G} \times G_m$$

$$L(N_{\hat{G}} / \hat{G}') = \text{Par}^1_{G,t} \quad ,$$

$$L(\text{St}^1_{\hat{G}'}) \simeq \text{Par}^1_{B,t} \times_{\text{Par}^1_{G,t}} \text{Par}^1_{B,t} \quad A \in \text{Alg}(Pr)$$

$$\text{HH}(A) \in \text{Alg}(\text{Vect}_{\overline{\mathcal{O}_E}}) : \left[\text{Vect}_{\overline{\mathcal{O}_E}} \rightarrow A \otimes_{A \otimes A^{\text{rev}}} A \rightarrow \text{Vect}_{\overline{\mathcal{O}_E}} \right]$$

$$\overline{\mathcal{O}_E} \xrightarrow{\quad} \text{HH}(A)$$

$$\text{Tr}(A\text{-mod}) \xrightarrow{\quad} Pr \quad , \quad A \mapsto A \otimes_{A \otimes A^{\text{rev}}} A$$

$$[\] : A\text{-mod} \rightarrow \text{Tr}(A\text{-mod})$$

$$\text{tr}(A) = \text{HH}(A) = \text{End}_{\text{Tr}(A)}([A^{\text{reg}}])$$

$f : X \rightarrow Y$ proper surj. rel. cotangent complex is perfect in degrees $[0, 1]$

$$Z = X \times_Y X \quad \left[X = N_{\hat{B}} / \hat{B}', \quad Y = N_{\hat{G}} / \hat{G}', \quad Z = \text{St}^1_{\hat{G}'} \right]$$

If Z affine, $\text{HH}(\mathcal{O}_{\text{Loh}}(Z)) = R\Gamma(LZ, \mathcal{O})$

$$LZ = LX \times_{LY} LX, \quad \text{End}_{LY}(Lf^* \mathcal{O}_{LX}) = \text{Hom}_{LY}(\mathcal{O}_{LX}, Lf^! Lf^* \mathcal{O}_X)$$

$$LZ \simeq LX \times_{LY} LX \xrightarrow{L\tilde{f}_2} LX \quad = \text{Hom}_{LX}(\mathcal{O}_{LX}, L\tilde{f}_x L\tilde{f}_! \mathcal{O}_X)$$

$$\begin{array}{ccc} L\tilde{f}_1 \downarrow & & \downarrow Lf \\ LX & \xrightarrow{Lf} & LY \end{array} \quad = \text{Hom}_{LZ}(\mathcal{O}_{LZ}, L\tilde{f}_! \mathcal{O}_{LX})$$

in any situation $\mathcal{O}_{LX} = \omega_{LX}$, $\mathcal{O}_{LZ} = \omega_{LZ}$

$$= R\Gamma_{LZ}(\mathcal{O}_{LZ}) = HH_*$$

(Hard) Th (Ben-Zvi - Nadler - Petyet '17 + ϵ)

$$\mathrm{Tr}(\mathrm{IndCoh}(Z)\text{-mod}) = \mathrm{IndCoh}(\underbrace{LY}_{\Delta})$$

↖ singular support condition

$$[\mathrm{IndCoh}(Z)] = Lf_* \mathcal{O}_X$$

$$\Rightarrow HH(\mathrm{IndCoh}(Z)) = \mathrm{End}(Lf_* \mathcal{O}_X)$$

$$[\mathrm{IndCoh}(X)] = \mathcal{O}_{LY}$$

Apply with $X = N\hat{B}/\hat{B}'$, $Y = N\hat{G}/\hat{G}'$

$$[\mathrm{IndCoh}(St)] = S_{G,t}^\perp, \quad \mathrm{Tr}(\mathrm{IndCoh}(St)) = \mathrm{IndCoh}(\mathrm{Par}_{\hat{G}}^\perp)$$

$$H_{G,t}^{\mathrm{alt}} = HH(\mathrm{IndCoh}(St)) = \mathrm{End}(S_{G,t}^\perp)$$

Compatibility w/ Whittaker data comes from

$K_0(N\hat{B}/\hat{B}') \hookrightarrow K_0(St_{\hat{G}}')$ - module is anti-spherical Hecke module

→

$$G = GL_2, \quad T \subset B \subset G$$

$$\begin{array}{ccc} (a_1, a_2) \curvearrowright \left| \begin{array}{c} \hookrightarrow T \\ \mathrm{Par}_T^\perp \end{array} \right. & \xleftarrow{2} & \begin{array}{c} \hline 1 \downarrow \hookrightarrow B \\ \hookrightarrow T \\ \hline \mathrm{Par}_B^\perp \end{array} \end{array}$$

$(Pa_a), N = \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$

$z=1$

↓ p

$$\begin{array}{c} \hline (1,1) \hookrightarrow \hat{G} \\ \hline \mathrm{Par}_{\hat{G}}^\perp \end{array} \quad (1,1) \hookrightarrow \hat{B}$$

Page 1

$$S_G^1 = P_* \mathcal{O}_{\text{Par}_B^1} = \mathcal{O}_{\text{Par}_G^1} \oplus \mathcal{O}_{\text{Par}_G^{\text{un}}}$$

$$\text{CLLC} \quad I_{B^{\text{op}}}^G \chi^{\text{un}} \longrightarrow S_G^1$$

is

$$\text{Ind}_I^G 1$$

is

$$\text{ind}_K^G 1 \oplus \underbrace{\text{ind}_K^G \overline{st}}_{W_{U,Y}}$$

$$\Rightarrow \text{CLLC} (\text{ind}_K^G 1) \longrightarrow \mathcal{O}_{\text{Par}_G^{\text{un}}}$$

$$\text{Hom} (\text{ind}_K^G \overline{st}, \text{ind}_K^G 1) \quad \text{free over } \mathbb{Z} (H_G^{\text{aff}})$$

$$0 \rightarrow \text{ind}_K^G \overline{st} \rightarrow \text{ind}_K^G 1 \rightarrow \chi_{G_m}^{\text{un}} \cdot \det \rightarrow 0$$

$$\begin{array}{c} 0 \rightarrow I(\text{Par}_G^{\text{un}}) \rightarrow \text{degree } -1 \\ \hookrightarrow \mathcal{O}_{\text{Par}_G^1} \rightarrow \mathcal{O}_{\text{Par}_G^{\text{un}}} \rightarrow 0 \end{array}$$

$$I_{B^{\text{op}}}^G (\chi_a \otimes \chi_b) \rightarrow P_* q^* \mathcal{O}_{\text{Par}_T^1} (q_b)$$

$$I_{B^{\text{op}}}^G \chi_a \otimes \chi_b = \mathcal{O}_{\text{Par}_G^1} (q_b) \quad \text{unless } a = qb$$